=== UNIT 3: CENTROID, CENTRE OF GRAVITY & MOMENT OF INERTIA ===

>> 3.1 Concepts of Centre of Gravity, Mass, and Centroid

In structural mechanics, structural stability and stress analysis fundamentally depend on cross-sectional geometric properties:
* Centre of Gravity (CG): The unique physical point through which the resultant gravitational body force (weight) of a body acts, regardless of orientation in a uniform gravitational field: x_cg = Integral(x dW) / W.
* Centre of Mass: The point representing the mean location of mass distribution: x_cm = Integral(x dm) / M. In uniform gravitational fields, CG coincides identically with CM.
* Centroid: The geometric center of a geometric figure (line, area, or volume). For plane areas, it depends solely on boundary geometry: x_bar = Integral(x dA) / A, y_bar = Integral(y dA) / A.

>> 3.2 Centroid of Composite Areas and First Moment of Area

The first moment of area Q about an axis represents the sum of area elements multiplied by their distances from that axis: Q_y = Integral(x dA) = A * x_bar; Q_x = Integral(y dA) = A * y_bar.
* Discrete Formulation for Composite Cross-Sections (I-beams, T-sections, Channels):
x_bar = (A1*x1 + A2*x2 + ... + An*xn) / (A1 + A2 + ... + An) = Sigma(Ai * xi) / Sigma(Ai).
y_bar = (A1*y1 + A2*y2 + ... + An*yn) / (A1 + A2 + ... + An) = Sigma(Ai * yi) / Sigma(Ai).
* Symmetry Properties: If a cross-section has an axis of symmetry, the centroid MUST lie on that axis. If it has two orthogonal axes of symmetry, the centroid is uniquely positioned at their intersection.

>> 3.3 Theorems of Pappus-Guldinus

Pappus-Guldinus theorems calculate surface areas and volumes of bodies of revolution generated by revolving plane curves and areas about non-intersecting coplanar axes:
* First Theorem (Surface Area): The surface area A generated by revolving a plane curve of length L about a coplanar non-intersecting axis equals the product of curve length and distance traversed by its centroid:
Area = 2 * pi * y_bar * L.
* Second Theorem (Volume): The volume V generated by revolving a plane area A about a coplanar non-intersecting axis equals the product of the area and distance traversed by its centroid:
Volume = 2 * pi * y_bar * A.

>> 3.4 Second Moment of Area (Moment of Inertia) and Radius of Gyration

Moment of inertia (MOI) quantifies resistance of a structural cross-section to bending and deflection under transverse loading:
* Area Moment of Inertia: I_xx = Integral(y^2 dA), and I_yy = Integral(x^2 dA).
* Polar Moment of Inertia (J or I_z): Resistance to torsional twisting: J = Integral(r^2 dA) = I_xx + I_yy.
* Radius of Gyration (k): Distance from reference axis where total area could be concentrated to yield the identical moment of inertia: k = sqrt(I / A) => I = A * k^2.

>> 3.5 Parallel and Perpendicular Axis Theorems

* Parallel Axis Theorem (Steiner's Theorem): The moment of inertia of any plane area about an arbitrary axis parallel to its centroidal axis equals the centroidal moment of inertia plus the product of total area and square of perpendicular distance between axes:
I_axis = I_centroidal + A * d^2.
* Perpendicular Axis Theorem: For any flat plane area lying in the x-y plane, the polar moment of inertia about the perpendicular z-axis equals the sum of moments of inertia about the coplanar x and y axes: I_zz = I_xx + I_yy.

>> 3.6 Centroidal Moments of Inertia for Standard Geometric Shapes

* Rectangle (base b, height h): Centroidal I_xx = (b * h^3) / 12; Base I_base = (b * h^3) / 3.
* Triangle (base b, height h): Centroidal I_xx = (b * h^3) / 36; Base I_base = (b * h^3) / 12.
* Circle (radius R, diameter d): Centroidal I_xx = I_yy = (pi * R^4) / 4 = (pi * d^4) / 64; Polar J = (pi * d^4) / 32.
* Semi-circle (radius R): Centroid y_bar = 4*R / (3*pi); Centroidal I_xx = 0.1098 * R^4.

[EXAMPLE 1: Centroid of an Asymmetrical Structural T-Section]

Problem Context: A structural steel T-section has a top flange of width b_f = 200 mm and thickness t_f = 20 mm, and a vertical web of height h_w = 180 mm and thickness t_w = 20 mm. Determine the location of the centroid (x_bar, y_bar) measured from the bottom center of the web.
Solution Step 1: Establish symmetry and sub-areas:

The section is symmetric about the vertical y-axis passing through web centerline => x_bar = 0.0 mm.
Subdivide into two simple rectangles:
- Element 1 (Flange): A1 = 200 * 20 = 4000 mm^2.
Centroid y1 measured from bottom = h_w + (t_f / 2) = 180 + 10 = 190 mm.
- Element 2 (Web): A2 = 20 * 180 = 3600 mm^2.
Centroid y2 measured from bottom = h_w / 2 = 180 / 2 = 90 mm.
Solution Step 2: Compute Total Area and First Moment of Area:
Total Area A = A1 + A2 = 4000 + 3600 = 7600 mm^2.
Sigma(Ai * yi) = (4000 * 190) + (3600 * 90) = 760,000 + 324,000 = 1,084,000 mm^3.
Solution Step 3: Compute Centroidal Height y_bar:
y_bar = Sigma(Ai * yi) / A = 1,084,000 / 7600 = 142.63 mm.
Conclusion: The centroid is located along the vertical line of symmetry at height y_bar = 142.63 mm from the bottom base.

[EXAMPLE 2: Surface Area and Volume of Torus via Pappus-Guldinus]

Problem Context: A circular cross-section of radius r = 50 mm is revolved about a non-intersecting coplanar axis at a distance R = 200 mm from the circle center to generate a circular ring torus. Using the Pappus-Guldinus theorems, calculate: (1) Total surface area of the torus, (2) Total volume enclosed.
Solution Step 1: Apply First Pappus-Guldinus Theorem for Surface Area:
Perimeter of generating circle: L = 2 * pi * r = 2 * pi * 50 = 314.16 mm.
Distance of curve centroid from axis of revolution: y_bar = R = 200 mm.
Distance traversed by centroid in one revolution: 2 * pi * y_bar = 2 * pi * 200 = 1256.64 mm.
Surface Area A = 2 * pi * R * L = (2 * pi * 200) * (2 * pi * 50) = 4 * pi^2 * R * r.
A = 4 * (3.14159)^2 * 200 * 50 = 39.4784 * 10,000 = 394,784 mm^2 = 0.395 m^2.
Solution Step 2: Apply Second Pappus-Guldinus Theorem for Volume:
Area of generating circle: A_circle = pi * r^2 = pi * (50)^2 = 7853.98 mm^2.
Volume V = 2 * pi * R * A_circle = (2 * pi * 200) * (pi * 50^2) = 2 * pi^2 * R * r^2.
V = 2 * (3.14159)^2 * 200 * (2500) = 19.7392 * 500,000 = 9,869,604 mm^3 = 0.00987 m^3.
Conclusion: Total surface area A = 0.395 m^2, Total enclosed volume V = 0.00987 m^3.

[EXAMPLE 3: Centroidal Moment of Inertia of a Built-up I-Beam]

Problem Context: An asymmetric structural I-beam has top flange 150 x 20 mm, web 20 x 260 mm, and bottom flange 250 x 20 mm. The centroid has been determined to be y_bar = 135 mm from the bottom base. Calculate the centroidal moment of inertia I_xx about the horizontal neutral axis.
Solution Step 1: Geometry and component centroids relative to base:
- Bottom Flange (1): b1 = 250, h1 = 20, A1 = 5000 mm^2; y1 = 10 mm. Distance d1 = |135 -

10| = 125 mm.
- Web (2): b2 = 20, h2 = 260, A2 = 5200 mm^2; y2 = 20 + 130 = 150 mm. Distance d2 = |135 - 150| = 15 mm.
- Top Flange (3): b3 = 150, h3 = 20, A3 = 3000 mm^2; y3 = 20 + 260 + 10 = 290 mm. Distance d3 = |135 - 290| = 155 mm.
Solution Step 2: Calculate centroidal MOI for each element (I_gi = b*h^3 / 12):
I_g1 = (250 * 20^3) / 12 = 166,667 mm^4.
I_g2 = (20 * 260^3) / 12 = 29,293,333 mm^4.
I_g3 = (150 * 20^3) / 12 = 100,000 mm^4.
Solution Step 3: Apply Parallel Axis Theorem (I_xx = Sigma [I_gi + Ai * di^2]):
I_xx1 = 166,667 + [5000 * 125^2] = 166,667 + 78,125,000 = 78,291,667 mm^4.
I_xx2 = 29,293,333 + [5200 * 15^2] = 29,293,333 + 1,170,000 = 30,463,333 mm^4.
I_xx3 = 100,000 + [3000 * 155^2] = 100,000 + 72,075,000 = 72,175,000 mm^4.
Total I_xx = 78,291,667 + 30,463,333 + 72,175,000 = 180,930,000 mm^4 = 1.809 * 10^8 mm^4.
Conclusion: Centroidal moment of inertia I_xx = 1.809 * 10^8 mm^4.

[EXAMPLE 4: Polar Moment of Inertia of Hollow Structural Column]

Problem Context: A hollow circular structural column has an outer diameter D = 300 mm and wall thickness t = 20 mm (inner diameter d = D - 2t = 260 mm). Determine: (1) Moment of inertia I_xx, (2) Polar moment of inertia J, (3) Radius of gyration k_xx.
Solution Step 1: Cross-Sectional Area A:
A = (pi / 4) * (D^2 - d^2) = (pi / 4) * (300^2 - 260^2) = (pi / 4) * (90,000 - 67,600) = (pi / 4) * 22,400 = 17,592.92 mm^2.
Solution Step 2: Centroidal Moment of Inertia I_xx:
I_xx = (pi / 64) * (D^4 - d^4) = (pi / 64) * (300^4 - 260^4) = (pi / 64) * (8.1 * 10^9 - 4.5698 * 10^9) = (pi / 64) * (3.5302 * 10^9) = 1.7329 * 10^8 mm^4.
Solution Step 3: Polar Moment of Inertia J (using perpendicular axis theorem J = I_xx + I_yy = 2*I_xx):
J = 2 * (1.7329 * 10^8) = 3.4658 * 10^8 mm^4.
Solution Step 4: Radius of Gyration k_xx:
k_xx = sqrt(I_xx / A) = sqrt(1.7329 * 10^8 / 17,592.92) = sqrt(9850.0) = 99.25 mm.
Conclusion: I_xx = 1.733 * 10^8 mm^4; J = 3.466 * 10^8 mm^4; k_xx = 99.25 mm.

>> 3.7 University Review & Examination Questions

* Q1: State and prove the Parallel Axis Theorem for the second moment of area.
Answer: Statement: The moment of inertia of any plane area about an axis parallel to a centroidal axis equals the centroidal moment of inertia plus the product of total area and the square of the distance between the two axes (I_AB = I_G + A*d^2). Proof: Let x-y be an axis system with origin at centroid G, and A-B be a parallel axis at vertical distance d. For an elemental area dA at coordinate y from G, its distance from A-B is y’

= y + d. I_AB = Integral((y + d)^2 dA) = Integral((y^2 + 2*y*d + d^2) dA) = Integral(y^2 dA) + 2*d*Integral(y dA) + d^2*Integral(dA). By definition: Integral(y^2 dA) = I_G; Integral(dA) = A; and Integral(y dA) = A*y_bar = 0 (since the origin is at the centroid). Thus: I_AB = I_G + A*d^2.

* Q2: Explain the significance of the radius of gyration in the design of structural compression members (columns).

Answer: The radius of gyration k = sqrt(I/A) measures how far from the centroidal axis the cross-sectional area is distributed. In civil engineering column design, the critical buckling load P_cr is governed by Euler's buckling formula: P_cr = (pi^2 * E * I) / L_eff^2 = (pi^2 * E * A) / (L_eff / k)^2. The ratio (L_eff / k) is known as the Slenderness Ratio (lambda). Columns always buckle about the axis having the minimum radius of gyration (k_min). Maximizing k for a given area A (e.g. using hollow tubes or I-sections) produces maximum buckling resistance.

>> 3.8 Key Takeaways and Revision Points

* Centroid is purely geometric; CG accounts for mass and gravitational distribution.
* First moment of area about centroidal axis is strictly zero (Integral y dA = 0).
* Pappus-Guldinus relates surface area to path of perimeter, and volume to path of area.
* Parallel Axis Theorem requires one axis to be the CENTROIDAL axis (I_axis = I_G + A*d^2).